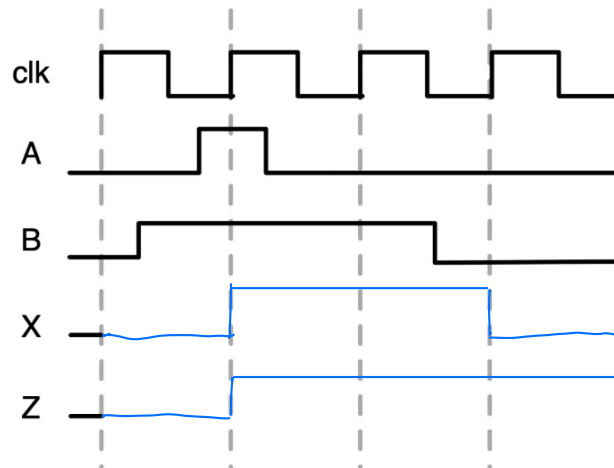


Question 1: SystemVerilog, Waveforms, and Schematics

a. Complete the timing diagram below and create a schematic equivalent to the SystemVerilog module below. Assume **X** and **Z** are low at the beginning of the diagram.

```
module dut (  
    input  logic A, B, clk, rst,  
    output logic X, Z  
);  
    logic tmp;  
  
    REGISTER_R #(.N(1)) delay_step0 (.clk(clk), .rst(rst), .d(B), .q(X));  
    REGISTER_R #(.N(1)) delay_step1 (.clk(clk), .rst(rst), .d(tmp), .q(Z));  
  
    assign tmp = (Z & X) | A;  
  
endmodule
```



b. Create a SystemVerilog module to represent the schematic and complete the timing diagram.

